

Harik Sodhi

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EDUCATION

University of Oxford (Christ Church) | *MEng Engineering Science* Oct 2025 – Jun 2029

- First-year: 92.4%, Distinction

AWARDS & HONOURS

- **Silver Medal, International Physics Olympiad** (2025) – 49th/420 competitors from 89 countries; highest-ranked competitor from Western Europe.
- **Winner, Berkeley Math Tournament Online Power Round** (2024) – First place in a proof-based competition on graph theory, cryptography, and computational complexity.
- **Ritangle National Champion** (2023) – Team Captain; 1st/1,200 teams in a national combinatorial optimisation challenge, developing and proving an optimal solution to a highly constrained scheduling problem with hundreds of interacting variables.

EXPERIENCE

Huawei UK R&D **Edinburgh**
Research Intern, Indoor Positioning Lab June 2026 – Present

- Contributing to research on wireless sensing and indoor localisation using WiFi signal measurements.
- Developing Python-based data analysis tools using Wireshark and OpenWrt to support experimental evaluation.

Oxford Alpha Fund **Oxford**
Quant Analyst & Director Jan 2026 – Present

- Selected through a competitive quantitative research and presentation process as a first-year undergraduate.
- Researched the question "Do BTC options systematically misprice volatility, and does this vary by day of the week?"; developed and backtested systematic trading strategies achieving a Sharpe ratio of 1.18 and a 70% win rate with a short iron butterfly trade. [Code]

Optiver Trading Academy **Oxford**
Participant Oct 2025 – Nov 2025

- Selected for Optiver's competitive Trading Academy programme.
- Traded ETFs, futures, and options on Optiver's proprietary simulation platform Optibook, applying market-making, delta-hedging, and tuning parameters while competing against other teams.

LEADERSHIP & ACTIVITIES

VEX Robotics & REC Foundation **London / Remote**
Global Student Advisory Board Member & Team Captain May 2021 – Mar 2024

- Selected as one of 12 students globally to advise the REC Foundation executive leadership on the future of the world's largest robotics competition.
- Served as Team Captain, Lead Programmer, and Lead Designer for 2 years; managed team operations and helped develop the competition robot and strategy.
- Developed autonomous programs in C++ using PID control and designed competition mechanisms in CAD.

PROJECTS

Market Making Simulator **Nov 2025**

- Developed a multi-agent market simulation modelling market makers, noise traders, and informed traders on a simple card game.
- Implemented strategy evaluation and performance analysis tools for comparing trading behaviour. [GitHub]

Rocket Lander **Apr 2026**

- Designed and developed a simulation of autonomous rocket landing dynamics, incorporating variable-mass propulsion, atmospheric drag, and gravity.
- Implemented RK4, a proportional control system, robustness testing, and constrained optimisation techniques to achieve stable autonomous touchdowns while reducing fuel consumption. [GitHub]

SKILLS

- **Programming:** Python, C++, MATLAB, LaTeX
- **Quantitative:** Statistical Modelling, Backtesting, Optimisation, Volatility Analysis, Numerical Methods
- **Libraries:** NumPy, Pandas, SciPy, PyTorch, Scikit-learn, Matplotlib
- **Tools:** Linux, Git, GitHub, SolidWorks (CSWE)